(4) Fisher's Ideal Index

$$P_{01}^{F} = \sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_0 q_1}} \times 100$$

$$= \sqrt{\frac{200}{160} \times \frac{130}{103}} \times 100$$

$$= \sqrt{1.578} \times 100 = 1.2561 \times 100$$

$$= 125.61$$

(5) Marshall-Edgeworth method

$$P_{01}^{ME} = \frac{\sum p_1 q_0 + \sum p_1 q_1}{\sum p_0 q_0 + \sum p_0 q_1} \times 100$$

$$= \frac{200 + 130}{160 + 103} \times 100 = \frac{330}{263} \times 100$$

$$= 125.48$$

Example 6.7

Calculate the price indices from the following data by applying (1) Laspeyre's method (2) Paasche's method and (3) Fisher ideal number by taking 2010 as the base year.

Commodity	2010		2011	
	Prices	Quantities	Prices	Quantities
A	20	10	25	13
B	50	8	60	7
C	35	7	40	6
D	25	5	35	4

Solution: Calculations

p_0	q_0	p_1	q_1	$p_0 q_0$	$p_0 q_1$	$p_1 q_0$	$p_1 q_1$
20	10	25	13	200	260	250	325
50	8	60	7	400	350	480	420
35	7	40	6	245	210	280	240
25	5	35	4	125	100	175	140
				970	920	1185	1125

12th_Statistics_EM_Unit_6.indd 164 1/28/2020 6:37:25 PM